

Network Design for All Star Gaming

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Abstract

This report documents the design, configuration, and testing of a network for All Star Gaming, a company operating across London and Manchester. The network was designed to support 50 employees across four office locations and a factory workshop. Cisco 2911 routers and Cisco 2960 switches were used and configured using Cisco Packet Tracer. A CIDR-based IP addressing scheme using VLSM was implemented to allocate subnets across all locations, and VLANs were configured to separate wired, wireless and factory traffic. RIP v2 was used to enable dynamic routing between sites. DHCP was configured to automate IP address allocation across all VLANs. Console password and local user authentication were applied to core devices. Connectivity testing confirmed successful inter-site communication between London and Manchester, along with verified VLAN and DHCP functionality. The report finishes with an evaluation of the design's performance, scalability, and limitations.

Introduction

All Star Gaming operates across London and Manchester, with around 50 employees split between four offices and a factory workshop. The business needs a network that keeps all of these locations connected, supports both wired and wireless access in the offices, and provides wireless connectivity for the factory floor, where machinery also needs to be networked.

Cisco Packet Tracer allowed for testing of this entire solution as well as verification of routing, switching, and security configuration without actual hardware. Cisco 2911 Routers function as the central router in each site (London & Manchester) to allow for communication between sites and to route local VLANs. Cisco 2960 Switches provide for wired network access to each connected device in each office. Each subnet has been assigned an appropriate size using VLSM and CIDR to meet the requirements of each location. Routing between locations is managed by RIP Version 2. VLANs prevent different types of traffic from being routed on the same segment. Address assignment is managed dynamically at the segment level through DHCP.

This report details the full design process, from planning and configuration through to connectivity testing and an assessment of where the design works well and where it could be improved.

Aim and Objectives

Aim: To design, configure, and validate a secure multi-site network for All Star Gaming that supports reliable communication between the London and Manchester offices.

Objectives:

- Design a network topology covering four office locations and a factory workshop.
- Develop an IP addressing plan using CIDR and VLSM.
- Configure VLANs to separate wired, wireless, and factory network traffic.
- Implement RIP version 2 to enable dynamic routing between sites.
- Configure DHCP to automate IP address allocation across all network segments.
- Apply password and SSH security to all core network devices.
- Verify network functionality through connectivity testing.

Network Design

All Star Gaming needed a network that could link four offices and a factory workshop across two cities, so the design had to account for both local connectivity within each site and reliable communication between London and Manchester.

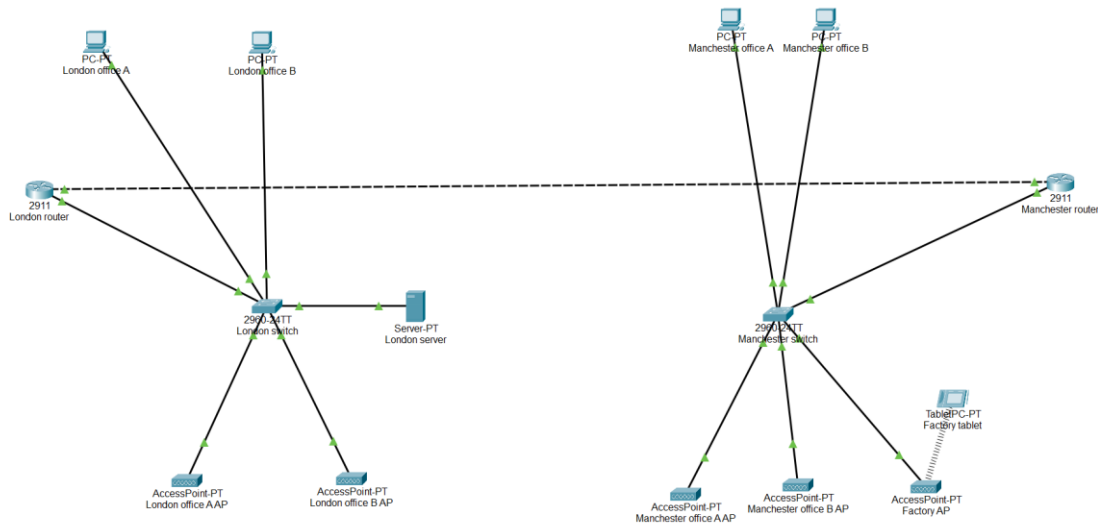
At the centre of every remote location is a Cisco 2911 Router. These were selected because they support sub-interface configuration on gigabit ethernet ports. This allows the same physical port to be used to route traffic in different VLANs, so no additional equipment is needed. Every router manages both local traffic as well as cross-city communications.

Cisco 2960 switches provided the connectivity of the wired equipment in each office. The Cisco 2960 supports VLAN tagging (802.1Q), meaning the switch can carry traffic for multiple VLANs simultaneously, with each access port assigned to a single VLAN and trunk ports carrying all VLANs between the switch and the router (Cisco, 2008). Access points are then connected to the switches to provide wireless coverage of all areas of both offices and the factory.

The physical connectivity to connect London and Manchester was implemented using fibre optic cable via HWIC-1GE-SFP modules in each router. This is because fibre optic can support a large amount of data (bandwidth) transmitted from one location to another over a long distance which makes it a perfect solution for connecting cities like London and Manchester.

A server was placed on the London side to function as a central resource for the organisation.

A 10.10.0.0/16 IP range was chosen as the base network for all of the offices. Using variable length subnet masking (VLSM) allows each of the subnets to be sized appropriately based on how many devices are actually going to connect in that specific location, as opposed to using the same number of usable IPs at each location. Every office will have its own /27 subnet; this will provide 30 usable IPs at each location. This is sufficient to cover the 20 devices per office and also has enough room for some potential future growth. Factory will use a /28 subnet because it requires less connectivity. This will allow us to make use of all 14 of the available usable IPs. If we were to use a /27 for factory, then we would waste the additional available address space by having no other devices needing connection. Lastly, we will be using a /30 for the WAN link between the two routers. This will give us exactly two usable addresses.



Location	VLAN	Type	Network Address	Prefix	Default gateway
London office A	10	Wired	10.10.10.0	/27	10.10.10.1
London office A	20	Wireless	10.10.10.32	/27	10.10.10.33
London office B	30	Wired	10.10.10.64	/27	10.10.10.65
London office B	40	Wireless	10.10.10.96	/27	10.10.10.97
Manchester office A	10	Wired	10.10.20.0	/27	10.10.20.1
Manchester office A	20	Wireless	10.10.20.32	/27	10.10.20.33
Manchester office B	30	Wired	10.10.20.64	/27	10.10.20.65
Manchester office B	40	Wireless	10.10.20.96	/27	10.10.20.97
Manchester factory	50	Wireless	10.10.20.128	/28	10.10.20.129
WAN Link			10.10.99.0	/30	10.10.99.1/2

VLANs

All the VLANs come back to the same physical interface on the 2911, so rather than buying a separate Layer 3 switch to manage routing between them, sub interfaces were configured instead. Each one maps to a specific VLAN and holds the default gateway address for that subnet, meaning devices in each

segment know where to send traffic that needs to go elsewhere. This approach is sometimes referred to as router on a stick, and for a network of this size it does the job without adding unnecessary hardware to the setup (Cisco, 2025).

```
London_Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	London_Office_A	active	Fa0/2, Fa0/4
20	London_Office_A_AP	active	Fa0/5
30	London_Office_B	active	Fa0/3
40	London_Office_B_AP	active	Fa0/6
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
London_Switch#
```

Copy

Paste

Inter-VLAN Routing

Because all VLANs terminate on the same physical router interface, sub interfaces were configured on the 2911 to manage routing between them. Each sub interface is assigned to a specific VLAN and given the default gateway address for that subnet. This is sometimes called router-on-a-stick and avoids the need for a separate Layer 3 switch (Cisco, 2025).

DHCP

DHCP was configured on both routers to automate IP address allocation for devices on each subnet. A separate pool was created for each VLAN, with the correct network range, default gateway, and DNS server specified. Gateway addresses were excluded from the pool to prevent them from being handed out to end devices (Cisco, 2012).

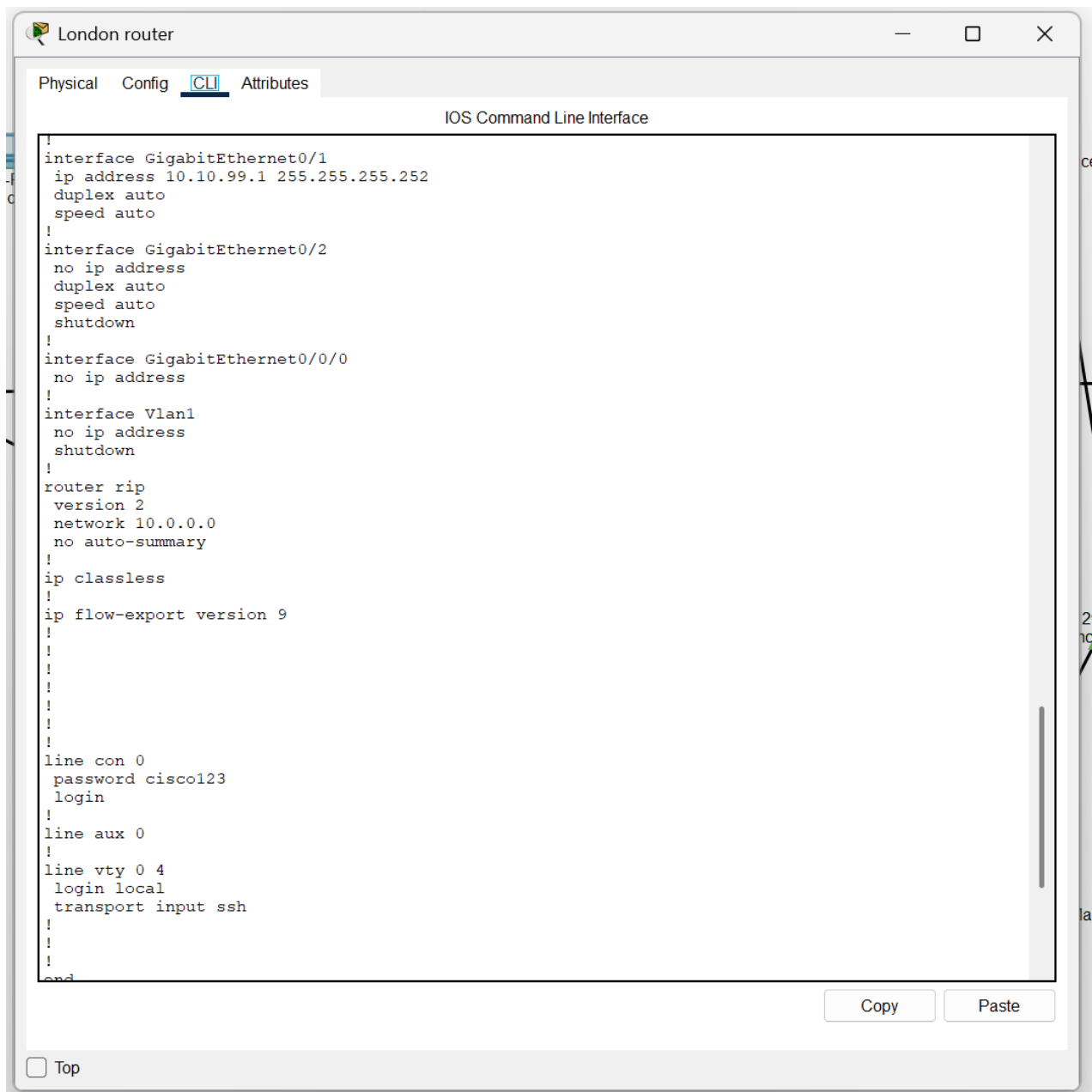
```

no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname London_Router
!
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX0lusIfka/
!
!
ip dhcp excluded-address 10.10.10.1 10.10.10.5
ip dhcp excluded-address 10.10.10.33 10.10.10.37
ip dhcp excluded-address 10.10.10.65 10.10.10.69
ip dhcp excluded-address 10.10.10.97 10.10.10.101
!
ip dhcp pool London_Office_A
network 10.10.10.0 255.255.255.224
default-router 10.10.10.1
dns-server 8.8.8.8
ip dhcp pool London_Office_A_AP
network 10.10.10.32 255.255.255.224
default-router 10.10.10.33
dns-server 8.8.8.8
ip dhcp pool London_Office_B
network 10.10.10.64 255.255.255.224
default-router 10.10.10.65
dns-server 8.8.8.8
ip dhcp pool London_Office_B_AP
network 10.10.10.96 255.255.255.224
default-router 10.10.10.97
dns-server 8.8.8.8
!
!
!
no ip cef
no ipv6 cef
!

```

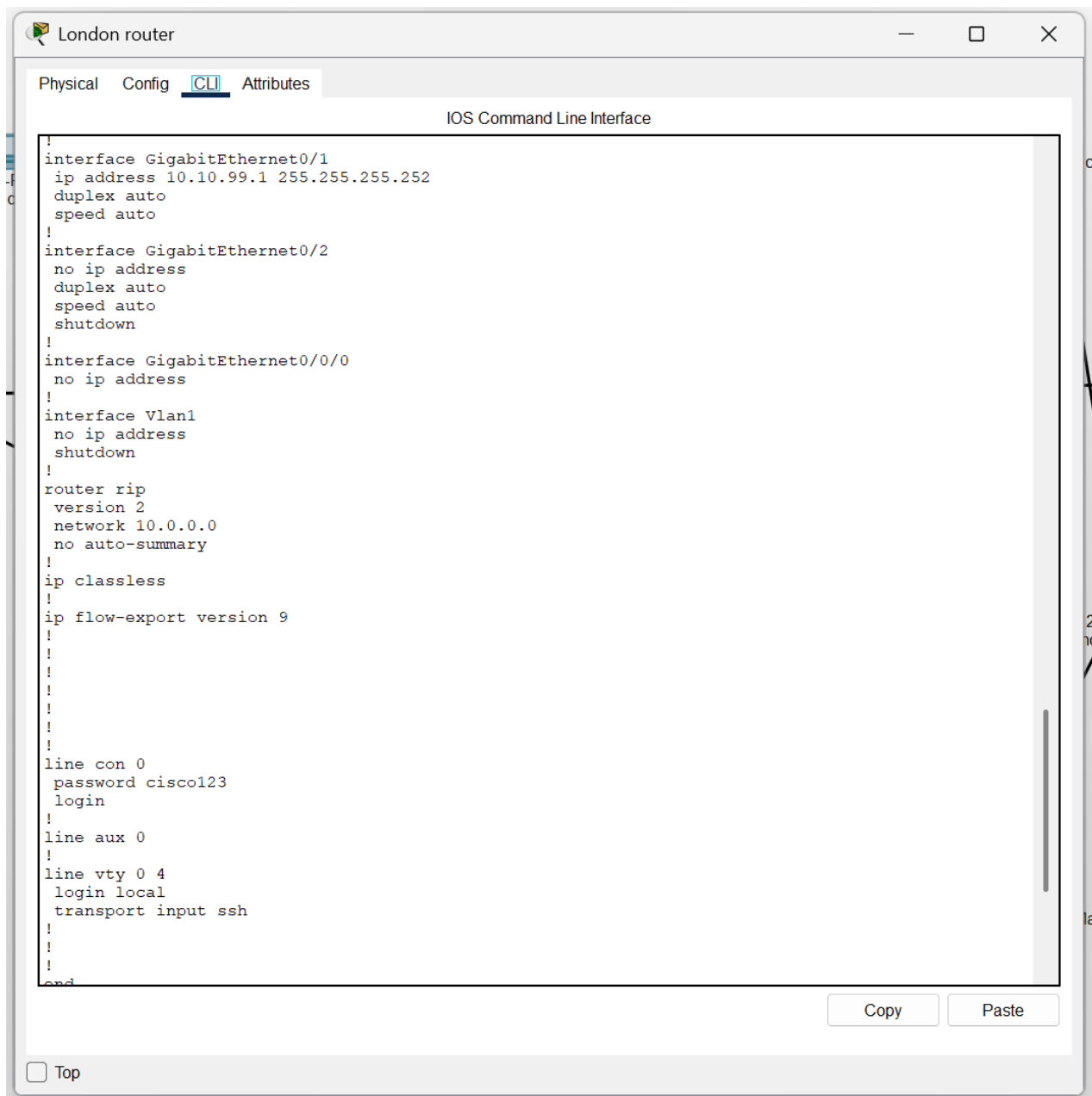
RIP Version 2

RIP version 2 was set up on both routers so they could share routing information with each other automatically. Every directly connected network was added into the RIP process, and auto-summary was turned off, which matters here because leaving it on would cause the router to advertise classful boundaries instead of the actual VLSM subnets being used. Version 2 was picked over version 1 specifically because it understands CIDR addressing, and without that the subnetting scheme across both sites simply would not work properly (Cisco, 2014).



SSH and password security

Console and enable passwords were configured on both routers to prevent unauthorised access to device management. SSH was enabled as the remote access method in place of Telnet, as SSH encrypts the management session, whereas Telnet sends credentials in plain text. A local username and password were created on each device to authenticate SSH sessions (Cisco, 2015).

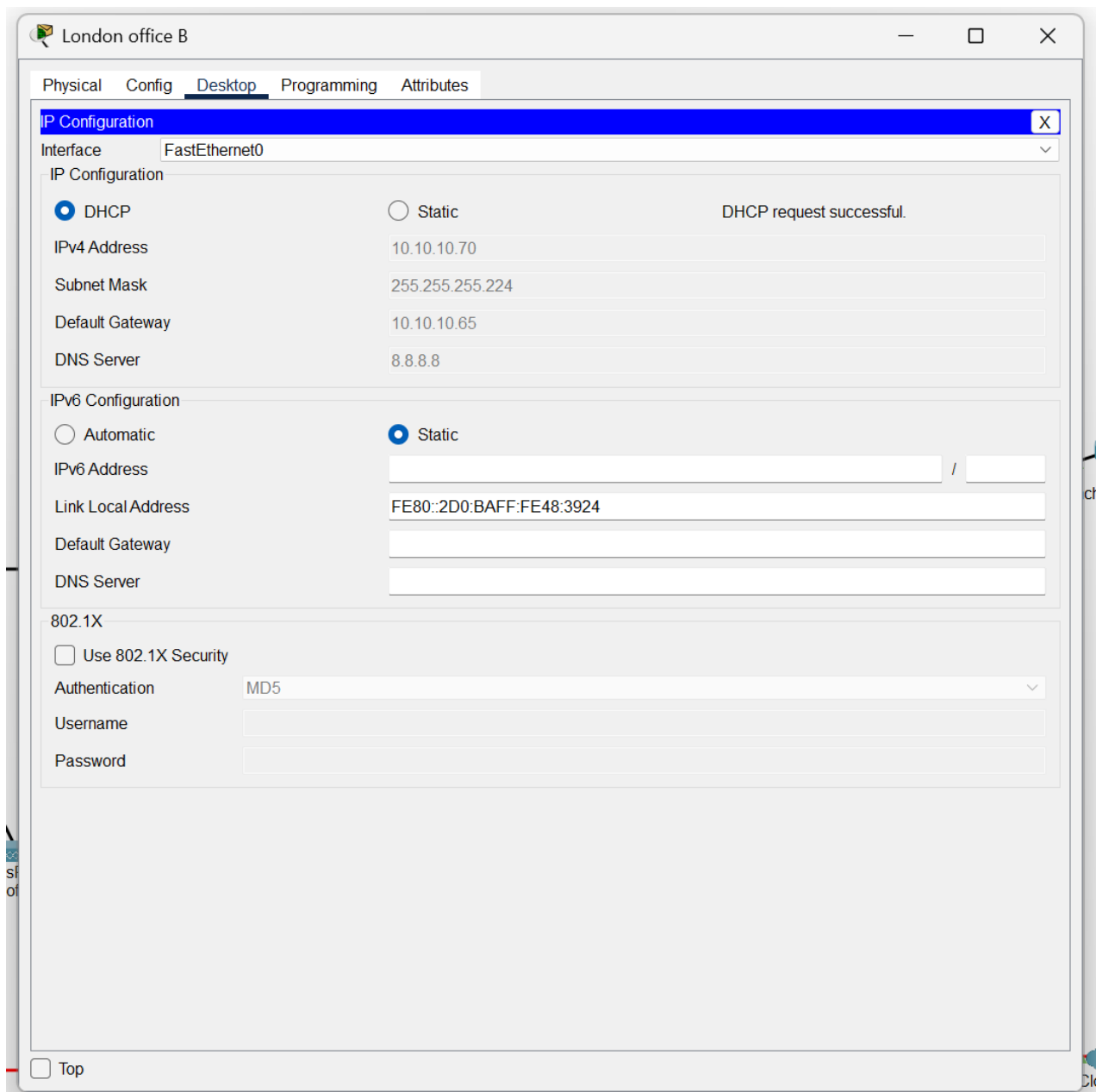


Network Configuration Tests

A series of tests was conducted to confirm that the network behaves as intended across all sites and that each of the design objectives has been met.

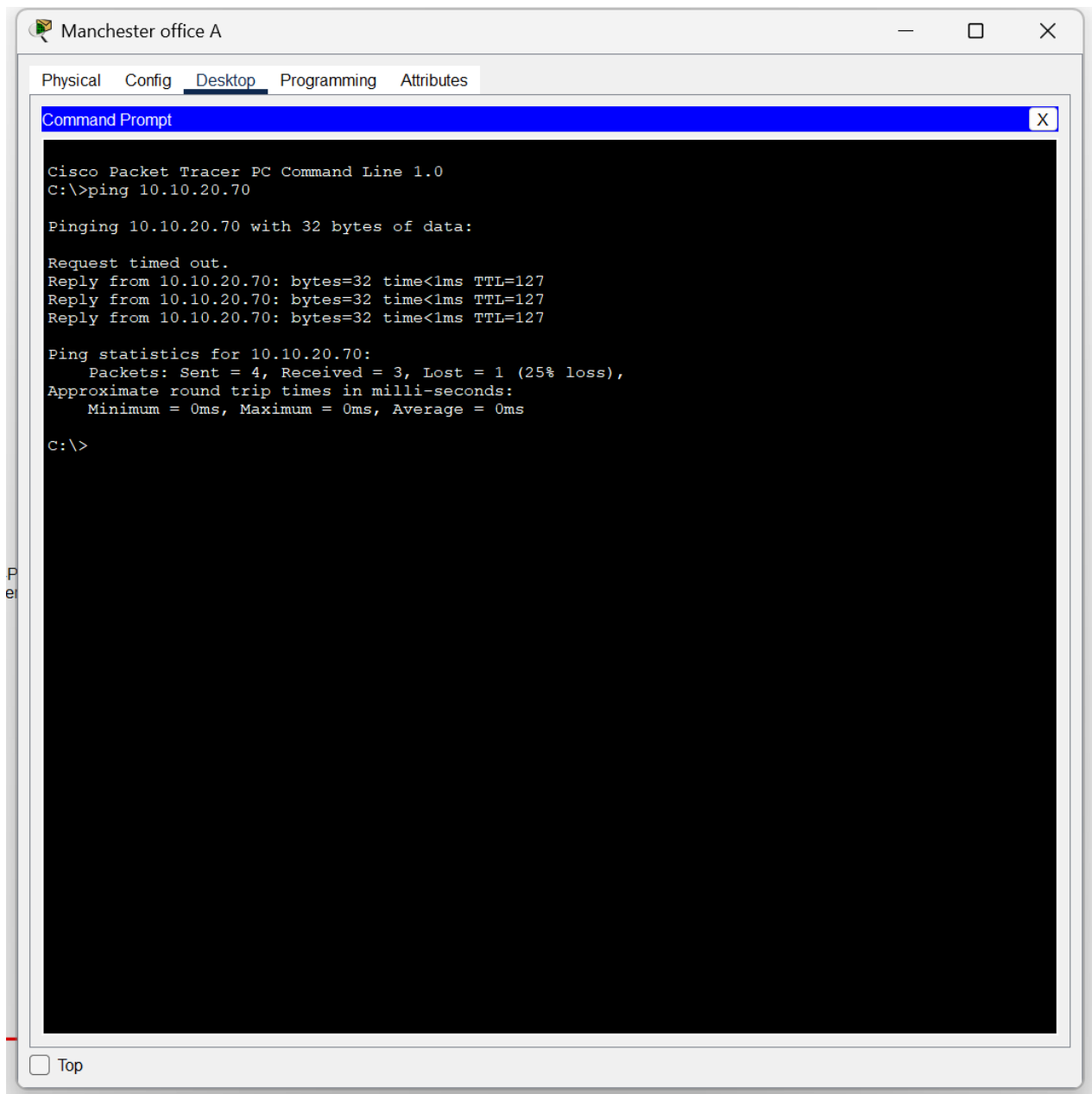
Test 1 — DHCP Verification

The London Office B (PC) had an automatic setting for the IP address. The `ipconfig` command returned 10.10.10.70 as the assigned IP address along with a Subnet Mask of 255.255.255.224 and a Default Gateway of 10.10.10.65. All of which matched the London Office B VLAN 30 Pool created by the London Router. This indicated that the DHCP Service functioned correctly. Prior to this implementation, each device would have to be assigned an IP address via manual input for each of the four offices and one factory location. Now devices could simply connect and receive their proper configuration eliminating the possibility of incorrect manual assignment due to typos or multiple conflicting assignments that may cause future issues.



Test 2 — Inter-VLAN Routing

A ping was sent from Manchester Office A to Manchester Office B. Both of these machines are located in separate VLANs on the same physical switch. This means a packet has to be sent off the switch, routed through a router to one of the other sub-interfaces of the router and then returned to the original source. The successful return of this ping indicates both sub-interfaces have been configured as they should have been and therefore the Router-On-A-Stick configuration is working properly.



Test 3 — Inter-Site Connectivity

A ping went from Manchester Office A to a PC in London Office A across the WAN link. Successful replies came back, confirming that the fibre optic connection between the two routers is passing traffic, RIP is distributing the correct routes on both sides, and devices in different cities can actually reach each other. This is the most important test in the set because without it, the two sites are effectively isolated from one another regardless of how well everything else is configured.

The screenshot shows the Manchester router's CLI interface. The tabs at the top are Physical, Config, CLI (selected), and Attributes. The main window displays the following text:

```
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Manchester_Router#
%SYS-5-CONFIG_I: Configured from console by console

Manchester_Router#
Manchester_Router#ping 10.10.99.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.99.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Manchester_Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Manchester_Router(config)#router rip
Manchester_Router(config-router)#network 10.10.99.0
Manchester_Router(config-router)#exit
Manchester_Router(config)#end
Manchester_Router#
%SYS-5-CONFIG_I: Configured from console by console

Manchester_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

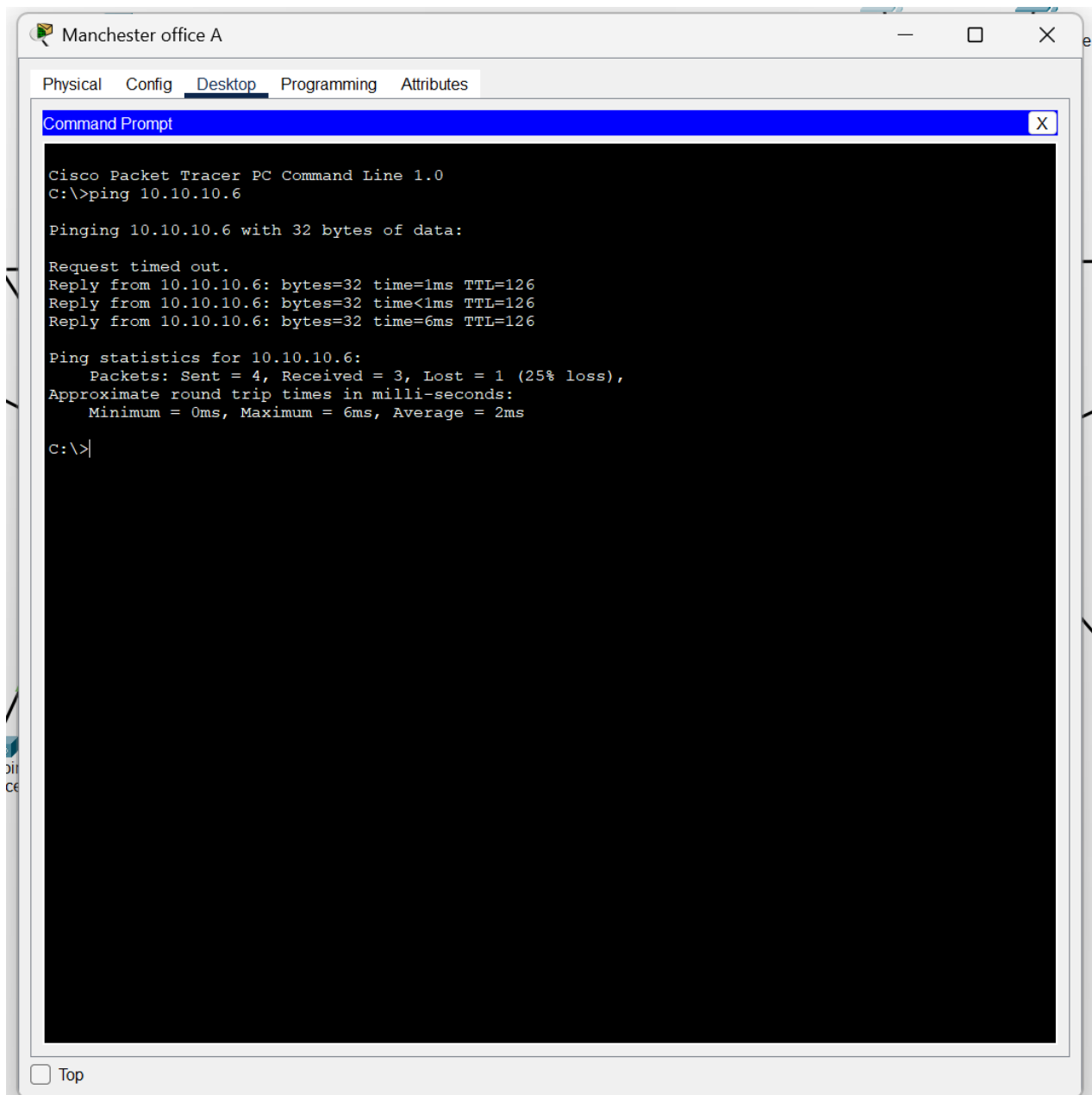
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 16 subnets, 4 masks
R       10.10.10.0/27 [120/1] via 10.10.99.1, 00:00:02, GigabitEthernet0/1
R       10.10.10.32/27 [120/1] via 10.10.99.1, 00:00:02, GigabitEthernet0/1
R       10.10.10.64/27 [120/1] via 10.10.99.1, 00:00:02, GigabitEthernet0/1
R       10.10.10.96/27 [120/1] via 10.10.99.1, 00:00:02, GigabitEthernet0/1
C       10.10.20.0/27 is directly connected, GigabitEthernet0/0.10
L       10.10.20.1/32 is directly connected, GigabitEthernet0/0.10
C       10.10.20.32/27 is directly connected, GigabitEthernet0/0.20
L       10.10.20.33/32 is directly connected, GigabitEthernet0/0.20
C       10.10.20.64/27 is directly connected, GigabitEthernet0/0.30
L       10.10.20.65/32 is directly connected, GigabitEthernet0/0.30
C       10.10.20.96/27 is directly connected, GigabitEthernet0/0.40
--More--
```

At the bottom of the CLI window, there are buttons for 'Copy' and 'Paste', and a 'Top' button with a checkbox.

Test 4 — RIP Route Verification

When you run "show ip route" on the Manchester router it pulls up the full routing table. The routes listed with an 'R' next to them are the important ones. They indicate routes which were not learned directly by this router but rather through the RIP protocol from the London router. All four addresses (10.10.10.0/27, 10.10.10.32/27, 10.10.10.64/27, and 10.10.10.96/27) appear in the routing table as RIP-learned routes via 10.10.99.1, confirming that the two routers are exchanging routing information automatically. Without the presence of those routes, the ping command in Test 3 would not be successful. The Manchester Router wouldn't know how to reach those destinations.



Network Evaluation and Discussion

The network delivers what All Star Gaming currently needs. Wired and wireless connectivity across four office locations, wireless coverage in the factory, and a working WAN link between London and Manchester. That said, there are still some limitations that need addressing.

The biggest structural weakness is that each site relies on a single router. If the London router goes down, everything at that site loses connectivity, including the link to Manchester. There is no redundant path in the design. Adding a second router at each site would resolve this, but for a company of 50 people, the added cost isn't justified right now. It is something worth revisiting as the business grows.

The same issue applies to the switches. One 2960 per site means a switch failure takes down all wired connectivity at that location. A second switch at each site would provide a backup, though again this doubles hardware costs. At this scale, the risk is manageable, but it is a known gap in the design.

RIP version 2 was chosen for inter-site routing because it is simple to configure and works well across a small two-site network. Its main limitation is the 15-hop count maximum, which would become a problem if the company expanded and added more locations. OSPF would be a more scalable replacement at that point, as it handles larger networks more efficiently and recovers from link failures faster than RIP does.

The VLSM addressing plan was sized with growth in mind. Each office subnet sits on a /27, which gives 30 usable addresses. With 20 devices per office right now, there are around 10 addresses spare in each subnet before anything needs restructuring. The 10.10.0.0/16 base range is large enough that adding new sites or departments further down the line would not require touching what is already configured. The factory subnet is a /28, so it is tighter, but given how few devices are connecting there it is the right call rather than handing out address space that would never get used.

The WAN link is a single connection between sites with no backup. If that link went down, London and Manchester would lose all communication until it was restored. A second WAN connection as a failover would fix this, even a basic broadband backup would keep the sites connected in the event of a primary link failure.

Security on the routers is managed through console passwords and local user authentication, which covers unauthorised device access. A logical next step would be to implement access control lists on the router interfaces to control which traffic is permitted between VLANs. This is straightforward to configure in IOS and would give the network administrator more control over how various parts of the network communicate with each other.

Overall, the design is proportionate to where All Star Gaming is now. The limitations identified are not flaws in the current implementation, but rather areas to address as the organisation scales up.

Conclusion

The multi-site network for All Star Gaming which is suitable for London and Manchester has been designed, configured, and successfully tested in this report. The wired and wireless network was installed for four offices and the factory workshop, which is exclusively wireless, with reliable communications between two cities.

The design implemented VLSM for efficient address space allocation between all locations, VLANs to segregate different types of traffic, and the dynamic routing of RIP version 2 to manage routing between sites automatically. The need for manual assignments of IPs was eliminated by the use of DHCP. Password authentication is used to secure the core devices only. Connectivity tests confirmed that all of these elements are working as intended.

There are areas where the design could be strengthened. Redundant hardware, a backup WAN link, and access control lists would all improve resilience and security as the business grows. These were not included in the current design due to the organisation's scale at this stage, but they represent a clear path for future development.

The network, as built, is functional, appropriately sized for All Star Gaming's current needs, and designed with sufficient address space and flexibility to support future expansion.

References

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Appendix A: